$M_{1229} = 2^{1229} - 1$

Factoring C284 — The 284-digit Composite Cofactor

Status: Project Paused

Date: 2026-10-04

1. Mersenne Number M_{1229}

$M_{1229} = 2^{1229} - 1$ is a Mersenne number with prime exponent 1229. Since 1229 is prime but does not appear in OEIS A000043 (the list of Mersenne-prime exponents), M_{1229} is composite. In fact, 1229 is the 187th term of OEIS A054723 — the sequence of prime exponents p for which M_p is composite. This sequence starts 11, 23, 29, 37, 41, 43, 47, 53, 59, 67, 71, 73, 79, 83, 97, 101, 103, 109, 113, ...

2. Complete Factorization (Verified)

Verified against three independent sources: Cunningham Project tables (Wagstaff), FactorDB, and Will Edgington's Mers database. The complete factorization is:

$M_{1229} = 36871 \times 46703 \times 10543179280661916121033 \times P55 \times C284$

Factor	Digits	Type	Discovery
36871	5	P (prime)	Trial division, early 20th century
46703	5	P (prime)	Trial division, early 20th century
10543179280661916121033	23	P (prime)	ECM, ~2000s
95362893553143925799960715151505830982[illegible]6033085857	55	P (prime)	ECM, pre-2018 (discoverer unknown)
C284 (284-digit composite)	284	C (composite)	UNFACTORED — still open

All four known prime factors satisfy the Mersenne-factor congruence $q \equiv 1 \pmod{2458} = 1 \pmod{2 \cdot 1229}$, and $q \equiv \pm 1 \pmod 8$ (the latter being required by quadratic reciprocity since 2 must be a quadratic residue mod q). The 284-digit cofactor C284 is currently the 4th-smallest open 'hole' in the Cunningham base-2 minus table (as of January 2, 2026).

2.1 The 4th-Smallest Cunningham Base-2 Hole

Rank	Mersenne Number	Cofactor Size	Status
1	M1207	c337	open
2	M1213	c297	open
3	M1217	c248	open
4	M1229	c284	open (THIS PROJECT)
5	M1231	c329	open

3. Structural Properties of C284

C284 has the following structural properties, all verified computationally:

Property	Value	Significance
Decimal digits	284	—
Bit length	943	—
C mod 4	3	Odd number of prime factors ≡ 3 (mod 4)
C mod 8	7	—
C mod 1229	1	—
C mod 2458	1	—
Perfect power?	No (checked k ⩽ 29)	—
Divides 2^1229 − 1?	Yes	Confirms Mersenne structure
Small factor (k ⩽ 40683)?	None	Trial division ruled out < 10^8

4. Work Done This Session

Three independent factoring methods were attempted during this session, using GMP-ECM 7.0.6 on a 2-core / 4GB RAM environment with approximately 25% CPU steal from the host hypervisor. Timing measurements were carefully calibrated against the 284-digit input, and the cumulative coverage from prior sessions was also taken into account when estimating the probability that an unknown factor of a given size would have been found.

4.1 Methods Attempted

Method	Curves/Attempts	Coverage	Result
ECM B1=1e7 (fast survey)	6	< 40 digits	No factor
ECM B1=1e8 (dedicated)	114	40-45 digits	No factor
P+1 (83 bases, B1=1e6/1e7)	83	Smooth (q+1)/2	No real factor
P-1 (B1=1e10, B2=1e15)	interrupted	Smooth (q-1)	Interrupted

4.2 Timing Benchmarks

B1	Stage 1	Stage 2	Total per curve	Curves/hour (2 cores)
1e7	13s	33s	47s	153
1e8	350s	143s	493s	14.6
1e10	~3400s	~1500s	~4900s	1.5
1e11	~34000s	~15000s	~49000s	0.15

4.3 GMP-ECM Expected Curve Counts (C284, 284 digits)

These are the published GMP-ECM 7.0.6 expected number of curves to find a factor of the given digit count (assuming one exists), with B1 = 10^8 and automatic B2 selection:

Digits	Curves Required
35	35
40	140
45	644
50	3,336
55	19,192
60	121,192
65	831,793
70	6,155,023
75	49,000,000
80	400,000,000

4.4 Cumulative Coverage

Combining this session's 114 curves at B1=1e8 with prior work (approximately 1200 curves at an effective B1 around 3.4e7 — the prior runs were limited by a 120-second timeout that prevented Stage 1 from reaching the nominal B1 of 3e12), the cumulative coverage for finding a factor of each size is approximately:

Factor Size	Curves Needed	Cumulative Coverage	Probability Excluded
< 40 digits	140	100+ (this session) + 1200 prior	~99%
40-45 digits	644	114 + ~1200 prior	~75%
45-50 digits	3,336	114 + minimal prior	~5%
50+ digits	19,192+	minimal	< 1%

Conclusion: The unknown prime factor(s) of C284 are almost certainly 50+ digits, and likely 70+ digits. This is consistent with the fact that no new factor of M_{1229} has been found since the P55 was discovered (pre-2018), despite ongoing ECM efforts by the Cunningham Project community.

5. Why We Stopped

The realistic computation time required to find a 70+ digit factor is infeasible on the available 2-core hardware. The following table shows the estimated total time at each B1 level, assuming 2 cores running in parallel at the measured rate:

B1	Target Factor Size	Total Time (2 cores)	Feasibility
1e10	55-60 digits	11 days	Marginally feasible
1e11	60-65 digits	177 days (6 months)	Not feasible in one session
1e12	65-70 digits	19 years	Infeasible
1e13	70-75 digits	~1,900 years	Infeasible

6. OEIS References

The Mersenne-number sequences in the On-Line Encyclopedia of Integer Sequences (OEIS) that are relevant to this project are summarized below. The primary sequence of interest is A054723, which lists the prime exponents p such that M_p is composite; 1229 is the 187th term of this sequence.

OEIS	Title	Relevance
A054723	Prime exponents p where 2^p - 1 is composite	Contains 1229 as term #187
A065341	Mersenne composites (the actual composite M_p values)	M_1229 is one of these
A244453	Prime factors of 2^A054723(n) - 1, ordered by bit-length	Contains factors of M_1229
A135980	Indices k into primes such that 2^prime(k) - 1 is composite	1229 = prime(203)
A000043	Mersenne-prime exponents (51-52 known)	Does NOT contain 1229
A001348	Mersenne numbers 2^p - 1 for prime p	Mix of prime and composite

7. External References

7.1 Cunningham Project (Sam Wagstaff, Purdue/CERIAS)

- Main tables: http://homes.cerias.purdue.edu/~ssw/cun/pmain126.txt (M_1229 at line ~945)
- Champions & First Five Holes (Jan 2 2026): http://homes.cerias.purdue.edu/~ssw/cun/champ.txt
- Notation guide: http://homes.cerias.purdue.edu/~ssw/cun/notat.txt
- Brillhart, Lehmer, Selfridge, Tuckerman, Wagstaff. Factorizations of b^n ± 1. Contemp. Math. 22, AMS, 3rd ed. 2002.

7.2 FactorDB & Mersenne Archives

- FactorDB M_1229 entry: http://factordb.com/index.php?query=2%5E1229-1
- Will Edgington's Mers database (Oct 2025): https://mers.sourceforge.io/mersenne.html
- GIMPS (Great Internet Mersenne Prime Search): https://www.mersenne.org/
- Wikipedia: https://en.wikipedia.org/wiki/Mersenne_prime

7.3 Mersenne Factorization Factory Paper

Kleinjung, Bos, Lenstra. "Mersenne factorization factory." ASIACRYPT 2014, eprint 2014/653. https://eprint.iacr.org/2014/653 — This batched-SNFS effort fully factored 17 Mersenne numbers with bit-lengths 1007–1199, the record being M1193. Since 1229 > 1199, M_1229 was NOT one of the seventeen, and the still-unfactored C284 confirms it has never been fully factored by SNFS.

8. Conclusion

The C284 cofactor of M_{1229} remains unfactored. It is one of the five smallest open 'holes' in the Cunningham base-2 minus table. Its unknown prime factor(s) are at least 50 digits, almost certainly 70+ digits, and may be substantially larger. Realistic paths to a future factorization include: (1) SNFS with significant infrastructure — only feasible beyond current SNFS reach (record is M1193); (2) distributed ECM with hundreds of cores running for years; (3) new mathematical breakthroughs such as large-scale quantum computing. For 2 cores and 4 GB of RAM, this problem is intractable in any reasonable timeframe, and the project is therefore paused.

9. Largest Known Prime Factor of M_{1229}

The largest known prime factor of M_{1229} is the 55-digit prime:

```
P55 = 9536289355314392579996071515160583098298616686033085857
```

This prime is proven prime (verified in FactorDB, status P). It was found by ECM before 2018, but the exact discoverer and date are not recorded in the dated Cunningham Project pages (1–148). It entered the cumulative Cunningham base-2 table from Will Edgington's Mers database / FactorDB, almost certainly found by an individual researcher running ECM. Its discovery is consistent with ECM being the only viable method for pulling a 55-digit prime out of a 370-digit composite cofactor (Pollard p−1 is infeasible since $P55 - 1 = 2^5 \cdot 7 \cdot 227 \cdot 1229 \cdot 1987 \cdot 36202654120499 \cdot 212136492363284166878443246266 1$ has its largest prime factor at 28 digits, requiring $B \approx 2 \times 10^{27}$).